INCH-POUND

MIL-E-53126(ME)
20 April 1992
(See 6.8)

MILITARY SPECIFICATION

ELECTRICAL DISTRIBUTION EQUIPMENT

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers five soldier-portable electrical distribution equipment end items (as well as auxiliary components) capable, singularly or in combination with each other, of being connected directly to a DOD generator set and supplying power to electrically operated equipment located inside shelters of various field unit complexes. Three of the end items are three phase systems, one is a single phase system, and one is a utility assembly containing extension cords, branch circuits and lighting equipment. The three phase systems can be used in various combinations with each other to create a small power network which is generally limited to a 100 feet radius per system (i.e. 300 feet total) in order to prevent excessive voltage loss. The utility assembly can be used with any of the systems to achieve internal shelter wiring and lighting. All three-phase distribution cables and the input cable to the 60-ampere single-phase system use MIL-C-3432 cable and MIL-C-22992, class L connectors. The other single-phase distribution cables use MIL-C-3432 cable and MIL-C-5015 connectors. Utility receptacles consist of hospital grade 120 volts "wall outlet" 20-ampere commercial receptacles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir, Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6150

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SPECIFICATIONS

FEDERAL

A-A-1898	- Cushioning Material, Packaging (Cellulosic, Water Absorbent).
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13226E7046	- Utility Assembly, Electrical, Receptacle and Lighting.
TA13227E6201	- Universal Adapter, 40/60 Amp, 3 Phase.
TA13229E6350	- Feeder System, Electrical, Three Phase, 200 Amperes/Phase.
TA13229E6351	- Feeder System, Electrical, Three Phase, 100 Amperes/Phase.
TA13229E6352	- Distribution System, Electrical, Single Phase, 60 Amperes.

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TA13229E6353 - Distribution System, Electrical, Three Phase, 40 Amperes/Phase.
13226E7035 - Branch Circuit Assembly, 8 Outlet, 120V, 20 A.
13226E7036 - Branch Circuit Assembly, 4 Outlet, 120V, 20 A.
13226E7045 - Light, Fluorescent, Single Tube.
13228E3356 - Cable Assembly, Transition, 5 Ton Van, Model M934-M935.
13229E6360 - Container, Storage, Reusable.

PUBLICATIONS

TM 9-6150-226-13 - Operator, Unit and Direct Support Maintenance Manual for Electrical Distribution Equipment.

(Copies of drawings and technical manuals required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FGS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The soldier-portable electrical distribution equipment (hereinafter called "distribution equipment") shall consist of auxiliary components (see 3.4) and the following five end items:

a. Utility assembly, electrical, receptacles and lighting, model M46, in accordance with TA13226E7046.
b. Distribution system, electrical, single phase, 60 amperes, model M60 A/P, in accordance with TA13229E6352.
c. Distribution system, electrical, three phase, 40 amperes/phase, model M40 A/P, in accordance with TA13229E6353.
d. Feeder system, electrical, three phase, 100 amperes/phase, model M100 A/P, in accordance with TA13229E6351.
e. Feeder system, electrical, three phase, 200 amperes/phase, model M200 A/P, in accordance with TA13229E6350.

When specified (see 6.2), items 3.1.a through 3.1.e shall be purchased separately.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of each of the five electrical distribution end items and auxiliary components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production items shall be made available, upon request, for inspection by the contracting officer or the contracting officer's designated representative.

3.2 First article. Unless otherwise specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. All material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The distribution equipment shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the distribution equipment may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.3.1 Cadmium plated fasteners. Cadmium plated fasteners shall not be used unless deemed absolutely necessary (see 6.6 and 6.6.1).

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the electrical distribution equipment may be newly fabricated from recovered materials to the maximum extent practicable, provided the electrical distribution equipment produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the electrical distribution equipment.

3.4 Auxiliary components. Listed below are auxiliary components authorized by Technical Manual TM 9-6150-226-13 to be separately requisitioned and used to supplement the end items as deemed necessary by the user. The reusable storage container, item af, drawing 13229E6360, is a basic hardwall container and may be issued to store other components/equipment. When required (see 6.2), any one or a combination of the auxiliary components shall be provided in accordance with the drawing(s) listed below:

Auxiliary Component Nomenclature	Drawing Number
a. Cable Assembly, Group PC, 60 Amp (4 Feet)	13226E7018
b. Cable Assembly, Group PC, 40 Amp/Phase (4 Feet)	13226E7019
c. Cable Assembly, Group PC, 100 Amp/Phase (4 Feet)	13226E7020
d. Cable Assembly, Group PC, 200 Amp/Phase (4 Feet)	13226E7021
e. Cable Assembly, Group S/F, 60 Amp (Dash No. 1 - 100 Feet)	13226E7022-1
f. Cable Assembly, Group S/F, 60 Amp (Dash No. 2 - 50 Feet)	13226E7022-2
g. Cable Assembly, Group S/F, 40/60 Amp/Phase (Dash No. 1 - 100 Feet)	13226E7023-1
h. Cable Assembly, Group S/F, 40/60 Amp/Phase (Dash No. 2 - 50 Feet)	13226E7023-2
i. Cable Assembly, Group S/F, 40/60 Amp/Phase (Dash No. 3 - 25 Feet)	13226E7023-3
j. Cable Assembly, Group S/F, 100 Amp/Phase (50 Feet)	13226E7024
k. Cable Assembly, Group S/F, 200 Amp/Phase (25 Feet)	13226E7025
l. Cable Assembly, Group E, 20 Amp (Dash No. 1 - 50 Feet)	13226E7032-1
m. Cable Assembly, Group E, 20 Amp (Dash No. 2 - 25 Feet)	13226E7032-2
n. Cable Assembly, Group E, 20 Amp (Dash No. 3 - 15 Feet)	13226E7032-3
o. Cable Assembly, Group E, 20 Amp (Dash No. 4 - 3.5 Feet)	13226E7032-4
p. Branch Circuit Assembly, 12 Outlet, 120V, 20 A (24 Feet)	13226E7034

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q.	Branch Circuit Assembly, 8 Outlet, 120V, 20 A (16 Feet)	13226E7035
r.	Branch Circuit Assembly, 4 Outlet, 120V, 20 A (8 Feet)	13226E7036
s.	Receptacle Group, 2 Duplex, Box Enclosure	13226E7040
t.	Rope Assembly, Branch Circuit Support, 53 Feet	13226E7041
u.	Light, Incandescent, Dual Bulb, Utility, 20 Foot Cord	13226E7043
v.	Securing Strap	13226E7044
w.	Light, Fluorescent, Single Tube (2 Foot Cord)	13226E7045
x.	Double Strap, Cable Carrier Assembly	13227E5821
y.	Single Strap, Cable Carrier Assembly	13227E5825
z.	Canister, Light Bulbs	13227E5829
aa.	Container, Transit and Storage	13227E5830
ab.	Universal Adapter, 40/60 Amp 3 phase	TA13227E6201
ac.	Cable, Assembly, Transition, 25 Outlet, Light Set	13228E3355
ad.	Cable Assembly, Transition 5 Ton Van, Model M934-M935	13228E3356
ae.	Container, Storage, Utility	13229E6362
af.	Container, Storage, Reusable	13229E6360

3.5 Power source compatibility. The feeder and distribution systems shall be suitable for connection to appropriate members of the DoD family of generator sets rated from 5kW through 200kW that operate at 120 volts single phase or 120/208 volts three phase at 50 HZ or 60 HZ. Each system shall be usable with 50 HZ or 60 HZ 120/208 volt commercial utility power when the proper interface is supplied and the commercial utility outlet is properly protected.

3.6 Environmental conditions. The distribution equipment shall conform to the following environmental requirements.

3.6.1 Operational conditions. When tested as specified in 4.5.2, the distribution equipment shall be capable of proving its intended function and operating without damage (see 6.4.1) under the following conditions:

a. Any ambient temperature from +120 to -25 °F. (Circuit breakers allowed no derating.)
b. Relative humidity up to 88 percent.
c. Rain (equipment normally external to shelter only, does not include model M46 assembly).

3.6.2 Storage and transit conditions. When tested as specified in 4.5.2, the distribution equipment shall not be damaged by storage (or operation after storage) under the following conditions:

a. Any ambient temperature from +160 to -28 °F.
b. Relative humidity up to 80 percent.

3.7 End item operational checkout. When tested as specified in 4.5.2.2, the distribution equipment shall be capable of providing its intended function and operating without damage.

3.8 Identification marking. The distribution equipment shall be identified in accordance with MIL-STD-130.

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3.9 Treatment and painting. The portions of the distribution equipment normally painted (see 3.9.1) shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals, chemical agent resistant coating, color green number 383, 3094 or tan 686A, 33446 conforming to MIL-C-46168. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.9.1 Unpainted surfaces. The extension cables, utility lights, loose connector, non-metal containers, branch circuits, ropes and power cable assemblies shall not be painted.

3.10 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

3.11 Human factors engineering. The distribution equipment shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to MIL-STD-1472, general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainer (5.9), and hazards and safety (5.13) as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Components and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable. As a minimum, every individual cable assembly shall be tested for continuity and proper wiring. In addition, every individual connector on every individual cable, feeder center, distribution center, and other assembly shall be mated and un-mated with a like connector several times to assure proper assembly and ease of function.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, inspection level II.

4.4.2 Quality conformance examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Quality conformance tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The distribution equipment shall be disassembled and examined (as specified in 4.3.1 for first article examination and 4.4.2 for quality conformance examination) for the defects listed in table I. Correct wire numbering and routing shall be verified for all circuits.

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TABLE I. Examination schedule (requirements).

No.	Defects	Requirement Paragraph
101.	Model M46 electrical assembly not as specified.	3.1
102.	Model M60 A/P distribution system not as specified.	3.1
103.	Model M40 A/P distribution system not as specified.	3.1
104.	Model M100 A/P feeder system not as specified.	3.1
105.	Model M200 A/P feeder system not as specified.	3.1
106.	Material not as specified.	3.3
107.	Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
108.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
109.	Contractor does not have documentation available for identification of material, material finishes or treatment.	3.3.3
110.	Use of cadmium plated fasteners not in accordance with the stated requirements.	3.3.3.1
111.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the electrical distribution equipment.	3.3.4
112.	Auxiliary components not as specified.	3.4
113.	Power source compatibility not as specified.	3.5
114.	Identification marking not as specified.	3.8
115.	Treatment and painting not as specified.	3.9
116.	Unpainted surfaces not as specified.	3.9.1
117.	Workmanship not as specified.	3.10
118.	Human factors engineering not as specified.	3.11

4.5.2 Tests.

4.5.2.1 Environmental. The distribution equipment shall be subjected to the test methods of MIL-STD-810 listed herein. Proper performance of intended functions and ability to sustain rated capacity for at least four continuous hours shall be verified during and immediately after each test method. Nonconformance to 3.6.1 and 3.6.2 shall constitute failure of this environmental test.

a. High temperature, method 501.2, procedure I (storage) and procedure II (operation). For procedure I, the relative humidity shall be maintained between one and seven percent at all times during each 24-hour storage cycle. The storage cycle shall be repeated for a total of seven 24-hour cycles and each cycle shall consist of: (1) maintaining the internal chamber temperature at 91 °F for two hours; (2) raising the temperature to 160 °F within nine hours and maintaining at 160 °F for one hour; and (3) lowering the temperature to 91 °F within 12 hours. For procedure II, the relative humidity shall be maintained between three and eight percent at all times during each 24-hour operation cycle. The operation cycle shall be repeated for a total of three 24-hour cycles and each cycle shall consist

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of: (1) maintaining the internal chamber temperature at 90 °F for two hours; (2) raising the temperature to 120 °F within eight hours and maintaining at 120 °F for four hours; and (3) lowering the temperature to 90 °F within 10 hours. The test item and all its components shall be operated at rated capacity for at least four hours during the time the temperature is at 120 °F.

b. Low temperature, method 502.2, procedures I (storage), II (operation), and III (manipulation). Operating personnel shall wear NBC (Nuclear Biological Chemical) protective or environmental (cold-weather) clothing during the low-temperature test to set up, operate, and disassemble the test item. For procedure I, the storage temperature shall be -28 °F and it shall be maintained for 24-hours. For procedure II, the operating temperature shall be -25 °F. For procedure III, option I shall be used. Procedure I shall be conducted first, procedure III second, and procedure II last. After step 4 of procedure I is completed, proceed to step 1 of procedure III.
c. Rain (equipment external to shelter only), method 506.2, procedure III (watertightness).
d. Humidity, method 507.2, procedure I (natural) and procedure II (induced). For procedure I, cycle 1 (hot-humid) shall be conducted using the 24-hour cycle 1 curve of figure 507.2-1. For procedure II, cycle 4 (hot-humid) shall be conducted using the 24-hour cycle 4 curve of figure 507.2-2. Cycles 1 and 4 shall be repeated as indicated in table 507.2-II for non-hazardous items.

4.5.2.2 End item operational checkout. The distribution equipment shall be operated at any ambient condition, as defined in 3.6.1. Nonconformance to 3.7 shall constitute failure of this test. During this checkout procedure every input circuit breaker shall be operated at its rated load for at least four continuous hours and every output circuit breaker shall be operated at its rated load for at least two continuous hours. Every cable, receptacle box or other component of the end item shall be utilized and operated at its rated capacity for at least one continuous hour during this checkout procedure. All switches and circuit breakers shall be turned on and off under rated load at least twice to verify proper functioning.

4.6 Inspection of packaging.

4.6.1 First article packaging inspection. The first article packages shall be examined for the defects listed in 4.6.2.3. The presence of one or more defects shall be cause for rejecting the first article pack.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product.

4.6.2.1.1 Electrical distribution equipment end item. For the purpose of inspection only, an end item unit of product shall consist of any complete end item covered by this specification that has been prepared for shipment.

4.6.2.1.2 Electrical distribution equipment auxiliary components. For the purpose of inspection only, a completed pack for a quantity of a single component, that has been prepared for shipment, shall be considered an auxiliary component

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unit of product for that component. A completed pack for a quantity of any other single component that has been prepared for shipment, shall be similarly considered.

4.6.2.2 Sampling. Sampling for examination shall be selected in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the major defects listed in table II. Presence of one or more defects shall be cause for rejection.

TABLE II. Examination schedule (requirements).

No.	Defects	Requirement paragraph		
		Level A	Level B	Level C
119.	Protective caps and covers not installed as specified.	5.2		
120.	Contact preservative used where not permitted.	5.2		
	PRESERVATION			
	End Items			
121.	Ends of cables not protected as specified.	5.3.1.1.1		
122.	Cables not coiled as specified.	5.3.1.1.1		
123.	Loose receptacles not preserved as specified.	5.3.1.1.2		
124.	Connectors not preserved as specified.	5.3.1.1.3		
125.	Rope assemblies not coiled and preserved as specifies.	5.3.1.1.4		
126.	Branch circuit assembly end connectors and receptacles not protected as specified.	5.3.1.1.5		
127.	Branch circuit assembly not coiled as specified.	5.3.1.1.5		
128.	Light bulbs not prepared for shipment in manner specified.	5.3.1.1.6		
129.	Utility light not preserved as specified.	5.3.1.1.7		
130.	Protective cover anchoring chains not taped to connector as specified for the distribution center.	5.3.1.1.8		
131.	Distribution center lid not secured in closed position as specified.	5.3.1.1.8		
132.	Distribution center not covered with wrap or bag as specified.	5.3.1.1.8		
133.	Wrap or bag not taped in place on distribution center as specified.	5.3.1.1.8		

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TABLE II. Examination schedule (requirements). (Continued)

No.	Defects	Requirement paragraph		
		Level A	Level B	Level C
134.	Electrical assembly components not consolidated, cushioned and secured in the assigned storage and transit container as specified.	5.3.1.2.1		
135.	Single phase, 60 ampere distribution system components, excluding the distribution center not consolidated, cushioned and secured in the assigned storage and transit container as specified.	5.3.1.2.2.1		
136.	Three phase, 40 ampere distribution system components, excluding the distribution center, not consolidated, cushioned and secured in the assigned storage and transit container as specified.	5.3.1.2.2.2		
137.	Components not preserved as specified for level C preservation.			5.3.2
	Auxiliary components			
138.	Cable assemblies not preserved as specified.	5.3.1.3.1		
139.	Rope assemblies not preserved as specified.	5.3.1.3.2		
140.	Branch circuit assemblies not preserved as specified.	5.3.1.3.3		
141.	Light bulbs not preserved as specified.	5.3.1.3.4		
142.	Utility light not preserved as specified.	5.3.1.3.5		
143.	Applicable components not placed in intermediate containers as specified.	5.3.1.3.6		
144.	Unlike components placed in same intermediate container.	5.3.1.3.6		
145.	Gross weight of intermediate container not as specified.	5.3.1.3.6		
146.	Intermediate container not as specified.	5.3.1.3.6		
147.	Universal adapter not preserved and placed in container connector as specified for the distribution center.	5.3.1.3.7		
148.	Fluorescent light not preserved as specified.	5.3.1.3.8		
	PACKING			
	End items			
149.	Electrical assembly not packed as specified.	5.4.1.1.1	5.4.2	
150.	Shipping container not as specified.	5.4.1.1.2.1	5.4.2	
151.	Distribution centers not secured to the shipping container bottoms as specified.	5.4.1.1.2.2	5.4.2	

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TABLE II. Examination schedule (requirements). (Continued)

No.	Defects	Requirement paragraph		
		Level A	Level B	Level C
152.	Vertical wood partition not installed in shipping containers as specified.	5.4.1.1.2.1 5.4.1.1.2.2	5.4.2	
153.	Clearance around distribution centers not as specified.	5.4.1.1.2.1 5.4.1.1.2.2	5.4.2	
154.	Filled storage and transit container not blocked within its cell as required.	5.4.1.1.2.1 5.4.1.1.2.2	5.4.2	
155.	Shipping container closure and strapping not as required.	5.4.1.1.2.1 5.4.1.1.2.2	5.4.2	
	Feeder systems			
156.	Shipping containers for the feeder systems not as specified.	5.4.1.1.3.1 5.4.1.1.3.2	5.4.2	
157.	Feeder systems not secured to shipping container bottoms as specified.	5.4.1.1.3.1 5.4.1.1.3.2	5.4.2	
158.	Vertical wood partition not installed in shipping containers as specified.	5.4.1.1.3.1 5.4.1.1.3.2		
159.	Clearance around feeder systems not as specified.	5.4.1.1.3.1 5.4.1.1.3.2		
160.	Coils not cushioned and blocked within their cell as specified.	5.4.1.1.3.1 5.4.1.1.3.2	5.4.2	
161.	Shipping container closure and strapping not as specified.	5.4.1.1.3.1 5.4.1.1.3.2	5.4.2	
	Auxiliary components			
162.	Auxiliary components not packed as specified.	5.4.1.1.4	5.4.2	
163.	Shipping container not as specified.	5.4.1.1.4		
164.	Box closure and strapping not as specified.	5.4.1.1.4		
165.	Unlike components mixed in shipping container.	5.4.1.1.4		5.4.3
166.	Components not packaged as specified for level C.			5.4.3
167.	Marking illegible, missing, or incomplete.	5.5	5.5	5.5

5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for each item for examination within the time frame specified (see 6.2), to prove prior to the start of production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The first article packs may utilize either the first article models or production models. When the first article models are utilized, and the Government requests a

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comparison between the first article models and the production models, any preservation and packing shall be removed at no expense to the Government.

5.2 General. All connector protective caps and covers shall be secured in position on their companion connectors. Contact preservatives shall not be used on connectors or connector components.

5.3 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Electrical distribution equipment end item.

5.3.1.1.1 Cable assemblies and cable assembly extensions. Each cable assembly and cable assembly extension shall have their ends protected with barrier material conforming to MIL-B-121, type II, class 2, held in place with no less than one complete wrap of tape conforming to PPP-T-60, type IV. The wrap shall cover the end and extend back over the side a minimum of 2 inches beyond the connector or pigtail end. Additional tape shall be applied over the barrier material at the pigtail end to prevent unsheathed wire ends from spreading apart. Each cable shall then be coiled to a compact diameter, sufficient to prevent excessive bending of the cable. Each coil shall be tied or secured a minimum of three places equidistantly spaced around the circumference of the coil.

5.3.1.1.2 Loose receptacles. Each loose receptacle shall be preserved in accordance with MIL-P-116, method IC-2.

5.3.1.1.3 Loose connectors. Each loose connector shall be preserved in accordance with MIL-P-116, method IC-2.

5.3.1.1.4 Rope assemblies. Each rope assembly shall be coiled and secured in a tight, compact coil and then preserved in accordance with MIL-P-116, method IC-3.

5.3.1.1.5 Branch circuit assemblies. The connectors at the ends of the branches shall be protected with barrier material as specified in 5.3.1.1.1. The receptacles shall be protected in the same manner. Each circuit assembly shall then be coiled in the manner specified in 5.3.1.1.1 with additional ties as necessary to prevent movement of the receptacles.

5.3.1.1.6 Light bulbs. The light bulbs (3 blue bulbs and 3 white bulbs) shall be preserved and prepared for shipment, in pairs, in a rigid, reusable light bulb canister (drawing number 13227E5829) designed to accommodate two properly separated bulbs.

5.3.1.1.7 Utility light. The guards shall be removed from the sockets and individually wrapped in cushioning material which is taped in place on the guards. The cushioning material and tape shall be as specified in 5.3.1.1.6. The cord shall be coiled and secured as specified in 5.3.1.1.6. The cord shall be coiled and secured as specified in 5.3.1.1.1. Each coil and the two wrapped guards shall then be enclosed in a bag in accordance with MIL-P-116, method IC-3.1

5.3.1.1.8 Distribution centers. All protective cover anchoring chains shall be taped to the connector with tape conforming to PPP-T-60, type IV. The distribution center lid shall be secured in the closed position with the latches

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provided. The center shall then be covered with a wrap or bag constructed of plastic film in accordance with L-P-378, type I, class 1, with a minimum thickness of 0.004 inches. The wrap or bag shall be of such dimension to completely cover the top, sides and ends of the center with enough surplus to be tucked under the body overlapping itself a minimum of 3 inches. The film shall be taped in place at the overlap with tape conforming to PPP-T-60, type IV. The lower portion of the support frame shall not be covered with the film.

5.3.1.2 Consolidation.

5.3.1.2.1 Electrical assembly. All electrical assembly components for one electrical assembly, which have been preserved as specified in 5.3.1.1.1 through 5.3.1.1.7 as applicable, in the quantities specified on TA13226E7046, shall be placed in the assigned utility storage container. Cushioning material conforming to A-A-1898, grade 2, style and size optional, 1/4 inch thick, shall be utilized within the container to cushion the components from each other and to prevent movement. The container shall be secured in the closed position by the means provided.

5.3.1.2.2 Distribution systems.

5.3.1.2.2.1 Single phase 60 amperes. All components for one single-phase, 60-ampere system, except the distribution center, which have been preserved as specified in 5.3.1.1.1 through 5.3.1.1.7 as applicable, in the quantities specified on TA13229E6352, shall be placed in the assigned storage and transit container. Cushioning and container closure shall be as specified in 5.3.1.2.1.

5.3.1.2.2.2 Three phase 40 amperes. All components for one three-phase, 40-ampere system, except the distribution center, which have been preserved as specified in 5.3.1.1.1 thru 5.3.1.1.7 as applicable, in the quantities specified on TA13229E6353, shall be placed in the assigned storage and transit container. Cushioning and container closure shall be as specified in 5.3.1.2.1.

5.3.1.3 Electrical distribution equipment auxiliary components.

5.3.1.3.1 Cable assemblies. Cable assemblies shall be preserved as specified in 5.3.1.1.1. The coiled cable shall be placed in a bag and sealed in accordance with MIL-P-116, method IC-3.

5.3.1.3.2 Rope assemblies. Rope assemblies shall be preserved as specified in 5.3.1.1.4.

5.3.1.3.3 Branch circuit assemblies. Branch circuit assemblies shall be preserved as specified in 5.3.1.1.5. The coiled circuit shall be placed in a bag and sealed in accordance with MIL-P-116, method IC-3.

5.3.1.3.4 Light bulbs. Light bulbs shall be preserved as specified in 5.3.1.1.6.

5.3.1.3.5 Utility light. Utility lights shall be preserved as specified in 5.3.1.1.7.

5.3.1.3.6 Intermediate containers. The individual auxiliary components which have been preserved as specified in 5.3.1.3.1 thru 5.3.1.3.5 shall be placed in

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intermediate containers. The intermediate containers shall be close-fitting boxes conforming to PPP-B-636, type CF, class domestic, variety SW, grade 2000, style optional. Box closure shall be as specified in the appendix to the box specification. Like components only shall be placed in an intermediate container. Unless otherwise specified (see 6.2), the quantity of a component within an intermediate container shall be sufficient to result in a gross weight of not less than 45 pounds but not more than 50 pounds.

5.3.1.3.7 Universal adapter. The universal adapter shall be preserved as specified in 5.3.1.1.8 for the distribution center. Additionally, each adapter shall be placed in a unit container complying with the requirements of PPP-B-636, type CF, grade V3c, style optional. Cushioning and blocking shall be applied to protect the adapter and to prevent free movement within the container. Container closure shall be in accordance with method V of the appendix to PPP-B-636.

5.3.1.3.8 Fluorescent light. Two fluorescent lights shall be preserved in a rigid, reusable carrying case. The lights shall be supported and cushioned within the carrying case with expanded cellular type material such as poly-styrene, polyurethane, polyethylene, or others, in order to prevent movement and breakage.

SPECIAL NOTE: Solid fiberboard or corrugated fiberboard shall not be used to manufacture the carrying case.

5.3.2 Level C. All components included in the electrical distribution equipment shall be preserved in a manner to prevent deterioration and damage during storage and shipment from the contractor to the initial destination.

5.4 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.4.1 Level A.

5.4.1.1 Electrical distribution equipment end items.

5.4.1.1.1 Electrical assembly. Each electrical assembly (see TA13226E7046), preserved and placed in the utility storage container as specified in 5.3, shall be individually packed in a close-fitting box conforming to PPP-B-601, overseas type. Box closure shall be in accordance with the appendix thereto and the strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.4.1.1.2 Distribution systems.

5.4.1.1.2.1 Single phase 60 amperes. All components for one single-phase, 60-ampere system (see TA13229E6352), shall be packed together in a box conforming to PPP-B-601, overseas type. The distribution center, preserved as specified in 5.3, shall be secured to the bottom of the box at one end with mechanical fasteners. The fasteners shall be either of U bolt or J bolt construction, or any other suitable means, of such size and quantity to prevent the center from moving within the box. Skids or intermediate cleats shall be added to the bottom of the box to enable the center to be properly secured. The box shall be constructed with one wood partition which divides the box vertically into two cells. One cell shall be for the center and the other cell shall be for the balance of the system's components which are enclosed within the applicable storage and transit container.

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The inside dimensions of the box shall provide a minimum of 1 inch clearance between any interior box surface and the top, sides and ends of the center, which is secured on one side of the partition, while providing adequate space on the other side of the partition for the filled storage and transit container. The partition shall provide the necessary rigidity, through its material strength and its method of attachment to the box body, to prevent movement of the filled storage and transit container toward the distribution center. Blocking shall be provided, as necessary, to prevent movement of the filled storage and transit container toward the distribution center. Blocking shall be provided, as necessary, to prevent movement of the filled storage and transit container in other directions. Box closure and strapping shall be as specified in 5.4.1.1.1.

5.4.1.1.2.2 Three phase 40 amperes. All components for one three-phase 40 ampere system (see TA13229E6353), shall be packed as specified in 5.4.1.1.2.1.

5.4.1.1.3 Feeder system.

5.4.1.1.3.1 Three phase 100 amperes. All components for one three-phase, 100 amperes system (see TA13229E6351), shall be packed together in a box conforming to PPP-B-601, overseas type. The feeder system, preserved as specified in 5.3, shall be secured to the bottom of the box at one end as specified in 5.4.1.1.2.1. Alternatively, when specified (see 6.2), holes shall be punched or drilled in the metal skids of the frame as a means for attachment. Additional wood skids shall be added to the bottom of the box to enable the center to be properly secured. The box shall be constructed with one wood partition which divides the box vertically into two cells. One cell shall be for the center and the other cell shall be for the three coiled cables making up the balance of the system's components. One inch clearance shall be provided for the center as specified in 5.4.1.1.2.1 and the partition shall provide the necessary rigidity to prevent movement of the coils toward the distribution center. The coils shall be cushioned with cushioning material specified in 5.3.1.1.6 and blocking shall be provided, as necessary, to prevent movement of the coils in other directions. Box closure and strapping shall be as specified in 5.4.1.1.1.

5.4.1.1.3.2 Three phase 200 amperes. All components for one three-phase, 200 amperes system shall be packed as specified in 5.4.1.1.3.1 for the 100 ampere system except that the referenced drawing shall be TA13229E6350 and that five coiled cables are required.

5.4.1.1.4 Electrical distribution equipment auxiliary components. The electrical distribution equipment auxiliary components which have been preserved as specified in 5.3 and the 13229E6362 utility container, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Box closure shall be as specified in the box specification and the appendix thereto and strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. Like components only shall be packed in each box.

5.4.2 Level B. Packing shall be as specified for level A except any boxes shall be domestic type and strapping may be coated finish.

5.4.3 Level C. All of the electrical distribution equipment shall be packed to ensure carrier acceptance and safe delivery to destination at lowest ratings in accordance with Uniform Freight Classification Rules or National Motor Freight

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Classification Rules. Only like components shall be packed in a shipping container for the auxiliary components.

5.5 Marking. In addition to any special or identification markings required by the contract or purchase order (see 6.2), each container shall be marked in accordance with MIL-STD-129. Center of balance markings shall be applied as specified therein.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The distribution equipment will be employed to meet the power distribution and illumination requirements of combat support and combat service support units operating near the combat zone. The distribution equipment will be used to transmit electricity through a series of cables, distribution centers, and branch circuits from generator sets to electrically operated equipment located within the tents of using units.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a model M46 electrical utility kit, model M60 A/P 60-ampere distribution system, model M40 A/P 40-ampere distribution system, model M100 A/P 200-ampere feeder system is required (see 3.1).
d. When a first article is not required (see 3.2).
e. When an auxiliary component or components is required (see 3.4).
f. When first article packs are not required (see 5.1).
g. Time frame for submission of first article packs when required (see 5.1).
h. Level of preservation and packing required (see 5.3 and 5.4, respectively).
i. When gross weight of intermediate containers for additional authorized components, when required, shall be other than specified (see 5.3.1.3.6).
j. When attachment holes are required in metal skids of feeder system frames (see 5.4.1.1.3).
k. Any special marking requirements (see 5.5).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition. The following definition shall apply throughout this specification.

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6.4.1 Damage. Damage is discoloring, swelling, corrosion, melting, permanent setting and loss of resiliency of gaskets, binding of parts, deformation, loosening, breakage, cracking, change of fit of any component or part, or any condition resulting in malfunction of the distribution equipment or any component or part.

6.5 First article pack. Any changes or deviations of the production packs from the approved first article packs will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the distribution equipment in accordance with this specification.

6.6 Cadmium plated fasteners. The contracting officer should require the contractor to justify the use of cadmium plated fasteners in writing (see 3.3.3.1). Citing standard DoD or commercial practice will not be considered sufficient justification. A specific rationale is required for each specific/different application of cadmium plated fasteners.

6.6.1 Cadmium plated fasteners part numbers. In those instances where cadmium plated fasteners are required and specified (see 3.3.3.1), drawings and technical manuals shall show these cadmium plated fasteners as separate part numbers to preclude the inadvertent use of zinc coated fasteners.

6.7 Subject term (key word) listing.

Branch circuit
Cable, electrical
Distribution centers
Electricity
Generator sets

6.8 Supersession data. This specification supersedes PD 6150-0026, dated 28 May 1986.

Custodian:
Army - ME

CIVIL AGENCY COORDINATING ACTIVITY
JUS - FPI

Preparing activity:
Army - ME

Project 6150-A219

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3. DOCUMENT TITLE
Electrical Distribution Equipment

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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